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On the path to a universal model for crystallization kinetics: Interpretation of the Šesták-Berggren equation in terms of the nucleation-growth concept

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ABSTRACT

An explicit equivalence between two dominant crystallization models, the flexible empirical Šesták-Berggren model and the physically meaningful nucleation-growth Johnson-Mehl-Avrami-Kolmogorov model, was found and quantified. The equivalence was confirmed to hold under any experimental conditions – isothermal annealing, heating from the glassy state, and cooling of the melt; the equivalence also results in identical crystal growth predictions based on the kinetic description by either of the two models. Based on this equivalence, the description of the crystallization data by the Šesták-Berggren model can be interpreted in terms of the physically meaningful nucleation-growth kinetics. Moreover, the deviations from this equivalence can now be unambiguously quantified, and the associated distortions of the crystallization peak can be attributed to concrete physicochemical interpretations. As such, the first physically meaningful interpretation is given to the autocatalytic Šesták-Berggren model (otherwise an empirical, purely mathematic function) when used for the description of the crystallization from the glassy matrix.

1. Introduction

Glass-ceramics combine the advantageous properties of both glass and ceramics, resulting in versatile and high-performance materials. Due to their fine-tuned crystalline structure, which provides enhanced mechanical properties, glass-ceramics often exhibit exceptional strength, toughness, and durability, exceeding those of the corresponding glass [1–5]. In addition, glass-ceramics often exhibit superior resistance to thermal shock and expansion-related mechanical stresses, where the glass-to-crystal ratio can be used to tailor the value of the coefficient of thermal expansion (CTE) [6–10]. Particularly important are the glass-ceramics with near-zero CTE, which are dimensionally stable even during large temperature changes, being ideal for utilization in high-precision instruments or in environments with rapid/large temperature fluctuations [11–13]. Similarly, the glass-ceramics are more thermally stable (with regard to the long-term high-T annealing) and exhibit higher chemical and corrosion resistance compared to their glass counterparts [14–19].

Applications-wise, glass-ceramics are not traditional only as cookware and kitchenware (stovetop-safe and induction-compatible products), but they find their place also in the medical field, where biocompatible glass-ceramics are used for bone implants or dental crowns [20–24]. Moreover, glass-ceramics are used as substrates for circuit boards and other microelectronic components due to their excellent electrical insulation, thermal stability, and precision machining capabilities [25–27]. In aerospace engineering, glass-ceramics are used in heat-resistant tiles and structural components where durability and resistance to extreme conditions are essential [28–30]. Furthermore, their optical properties make them suitable for high-performance lenses, mirrors, and optical systems, including those used in advanced scientific instruments [31–33].

The key aspect for producing high-quality glass-ceramics with precisely defined properties is an accurate control of the crystal growth process – both from the microscopic point of view (defining the size of the crystallites and/or their morphology) and macroscopic point of view (regulating the overall degree of crystallinity). Out of the two, the latter is usually of higher importance. Mathematically, the macroscopic crystallization kinetics is standardly described by the solid-state kinetic equation [34]:

$$\frac{d\alpha}{dt} = K(T(t)) \bullet f(\alpha) \tag{1}$$

where $d\alpha \cdot dt^{-1}$ is the rate of conversion from the amorphous to the crystalline phase (with α being the degree of conversion ranging from 0 to 1, and t being time), K(T(t)) is the rate constant (generally dependent on the temperature T or, more precisely, on the temperature

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program T(t)), and f(α) is the kinetic model function (responsible for the asymmetry of the derivative crystallization signals).

For the crystallization proceeding during heating from the glassy state (the so-called cold crystallization), the Arrhenius equation is usually used for the rate constant [35]:

$$K(T(t)) = A \bullet e^{-E/RT} \tag{2}$$

where A is the pre-exponential factor (s^{-1}), E is the apparent activation energy of the macroscopic crystallization process (J·mol^{-1}), and R is the universal gas constant (J·mol·K^{-1}). However, as recently demonstrated [36–39], such an approximation does not always accurately represent the real physicochemical behavior of glassy materials, which can exhibit temperature-dependent E and/or A, leading to:

$$K(T(t)) = A(T) \bullet e^{-E(T)/RT} \tag{3}$$

The rate constant expressed in such a way does not necessarily have to increase exponentially with T (as if E and A are constants), but it can exhibit a maximum somewhere between the glass transition temperature (T_g) and melting temperature (T_m) [36]. Such behavior is a much better approximation of the truly observed nucleation and crystal growth behaviors [40,41]. Similar approach can also be used for the crystallization during cooling from the molten state (the so-called hot crystallization), where the most well-known expression for the rate constant is the Hoffman-Lauritzen theory [42,43] inspired by the polymer science:

$$K(T(t)) = A \bullet \exp\left(-\frac{U^*}{R(T(t) - T_\infty)}\right)\exp\left(-\frac{K_G}{T(t) \bullet \Delta T(t) \bullet g(T(t))}\right) \tag{4}$$

where A is a pre-exponential factor (essentially translating the manifestation of the exponential dependences to the macroscopic level), U* is the activation energy for the segmental movements of the polymer chains (considered to be 6300 J·mol^{-1} for the majority of polymeric materials [42]), R is the universal gas constant, T_∞ is the temperature under which all motions associated with a viscous flow must be significantly slower than the experimental time scale given for the crystal growth (customarily $T_\infty = T_g - 30$ °C), ΔT is the undercooling (defined as $\Delta T = T_m^{eq} - T$, where T_m^{eq} is the equilibrium melting temperature), g(T(t)) is the correction factor defined as $g = 2\,T/(T_m^{eq} + T)$, and K_G is the kinetic parameter (constant for certain temperature levels/regimes) associated with nucleation. Similarly to Eq. 3, also K(T) defined by Eq. 4 exhibits a maximum between T_g and T_m and can thus be considered a good approximation of the physicochemical reality.

Regarding the kinetic model function f(α), two equations are regularly utilized – the physically meaningful nucleation-growth Johnson-Mehl-Avrami-Kolmogorov, JMAK, model [44–48] (Eq. 5) and the empirical autocatalytic Šesták-Berggren, AC, model [49] (Eq. 6):

$$f(\alpha) = m(1-\alpha)[-\ln(1-\alpha)]^{1-(1/m)} \tag{5}$$

$$f(\alpha) = \alpha^M(1-\alpha)^N \tag{6}$$

where m, M, and N are the kinetic exponents of the two models. The JMAK kinetic exponent is usually interpreted in accordance with Table 1 [50], expressing the nucleation regimes and dimensionality of growing crystallites – the possibility of such interpretation is the main incentive for using the JMAK model.

However, the JMAK kinetics exhibits (regardless of the *m* value) a very strictly defined α-T (or α-t in the case of isothermal measurements) profile that is reflected into a specific asymmetry of the derivative kinetic peaks [51] – such as those measured by differential scanning calorimetry (DSC) or differential thermal analysis (DTA). Note that the peak asymmetry is defined as ratio between the distances from the peak onset to its center (numerator) and from its center to its endset (denominator); usually, the measurements are performed in 1/10 of the peak height – see the Supplemental online material for a visual depiction. The ratio equal to 1 defines a symmetric peak, the ratio > 1 denotes a fronting peak (negative asymmetry), the ratio < 1 corresponds to a tailing peak (positive asymmetry). The fingerprint JMAK asymmetry is shown in Fig. 1A, where the kinetic peaks are characteristically skewed to higher α values (higher T during heating, i.e., the negative asymmetry). However, very often, calorimetric crystallization data are obtained, which exhibit different asymmetry and thus cannot be accurately described in terms of the JMAK model. In such a case, the highly flexible AC model is used to achieve an accurate description of the data. Note that such a choice is inevitable because one cannot utilize any kinetic modeling where the fundamental description (fit of the experimental data by the mathematic equations) is not accurate. Despite being semi-empirical, the AC model can be interpreted at least in regard to its resemblance to the nth-order kinetic model [52], which is defined as:

$$f(\alpha) = (1-\alpha)^N \tag{7}$$

where the kinetic exponent N represents the reaction order. Note that the exponent in Eq. 7 can be written either as "n" or "N"; here, the capital letter is used to stress the resemblance with the similar exponent in Eq. 6. In this connotation, the AC model can be interpreted as an autocatalyzed nth-order model, where the kinetic exponent M defines the degree of autocatalytic behavior. The impact of the exponent N in Eq. 6 is identical to that known for the nth-order model, i.e., shifting the asymmetry from the negative skewing [49,52] (found for low N values) to the positive skewing (found for N > 1.5), while retaining the onset edge of the kinetic peak – see Fig. 1B. Effectively, the increasing N value leads to a slow-down of the kinetic process. For the kinetic exponent M, the interpretation is more complex since the effect of its increase is double – the kinetic peak shifts to higher T (i.e., the process slows down), but the peak is higher and narrower (i.e., the evolution of the crystallization heat is more rapid) – see Fig. 1C. Notably, in the case of the kinetic exponent M, also its negative values have their meaning. Nonetheless, since the physicochemical background of the nth-order model is not suitable for the interpretation of the nucleation and growth characteristics of the crystallization mechanisms, the AC model is generally perceived as empirical and/or without physical meaning for the crystallization processes.

The aim of the present study (based on theoretical simulations, but taking the kinetic parameters from the experimental crystal growth description of previously studied glassy and polymer materials) is to find the correlation between the JMAK and AC models, and to verify the universality of this correlation. Moreover, the interpretation of the AC kinetics in terms of the JMAK nucleation-growth concept will be developed and tested. The testing will be particularly focused on the validity of the kinetic predictions produced using a set of AC kinetic parameters deviated from those exactly corresponding to the JMAK concept.

Table 1
Physicochemical interpretations of the JMAK exponent *m* (third and fourth columns).

growth dimension	nucleation mechanism	reaction rate-controlled model	
		diffusion	interface
three-dimension	heterogeneous nucleation	1.5	3
	inhibited nucleation	1.5 – 2.5	3 – 4
	homogeneous nucleation	2.5	4
	accelerated nucleation	> 2.5	> 4
two-dimension	heterogeneous nucleation	1	2
	inhibited nucleation	1 – 2	2 – 3
	homogeneous nucleation	2	3
	accelerated nucleation	> 2	> 3
one-dimension	heterogeneous nucleation	0.5	1
	inhibited nucleation	0.5 – 1.5	1 – 2
	homogeneous nucleation	1.5	2
	accelerated nucleation	> 1.5	> 2

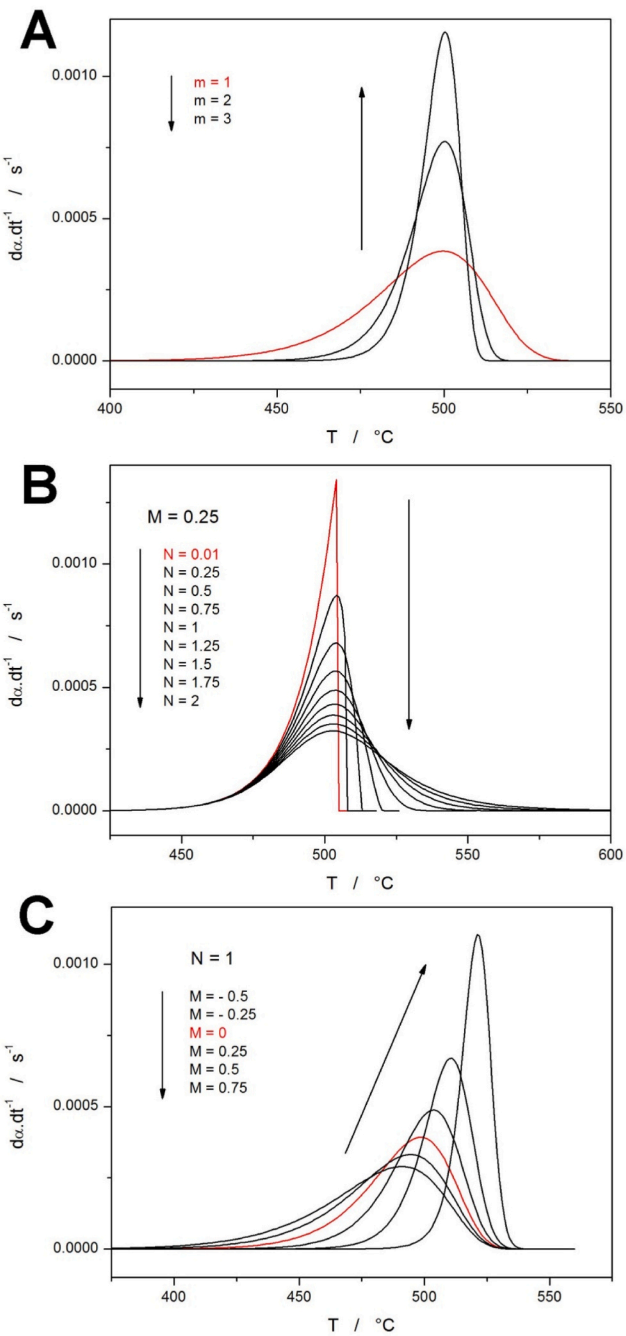

Fig. 1. A) Example derivative kinetic data (akin to, e.g., DSC or DTA crystallization signals) corresponding to the JMAK kinetics with defined values of the exponent *m*. B) Example AC kinetic peaks simulated for a fixed value of M and varying N. C) Example AC kinetic peaks simulated for a fixed value of N and varying M.

2. Results

To find the correlation between the JMAK and AC models, the JMAK kinetic peaks were simulated under different conditions (different temperature programs and different K(T) and m values from Eqs. 1 and 5) and fit by the AC model. The mathematical modeling based on Eqs. 1 and 6 utilized the Euler approach (with initial $\alpha_{init} = 0.0001$, and $\Delta t = 1$ s) [53] for the simulation part and the Levenberg-Marquardt algorithm for the non-linear optimization part. To achieve universally acceptable conclusions, the simulations of the JMAK kinetic peaks were performed in three batches, covering the majority of the temperature range relevant to the crystallization processes – see Fig. 2 (for each temperature range, several series of the JMAK kinetic peaks were simulated, as will be described further). The low-temperature batch (exhibiting the crystallization peaks near 50 °C) can represent the behavior of low-molecular organic glasses (e.g. pharmaceutical substances) and thermoplastics [54–58]; the middle-temperature batch (showing crystallization near 200 °C) can represent, e.g., chalcogenide, tellurite, phosphate or borate glasses, for which the crystallization can occur from 150 to 450 °C [59–62]; the high-temperature batch (showing crystallization near 500 °C) can represent, e.g., aluminosilicates, borosilicates, heavy-metal phosphates, fluorides, or certain soda-lime glasses, where the crystal growth often proceeds in the 500 – 700 °C range [63–67]. Note that in the present section, the JMAK-AC correlation will be shown to be explicitly universal, so the temperature ranges depicted in Fig. 2 can be arbitrarily extended.

The main batch of the tested JMAK data was simulated for the standard non-isothermal heating program, imitating the preparation of ceramics and glass-ceramics from the glassy material – Fig. 3A shows evolution of the shape of the JMAK kinetic peak with changing exponent m (simulated for the heating rate $q^+ = 1$ °C·min^{-1} and for the Arrhenian K(T) expressed by Eq. 2 with E = 300 kJ·mol^{-1} and $\ln(A/s^{-1}) = 39.8$). Note that the comparisons of the crystallization dα·dt^{-1}-T signals and K(T)-T dependencies are for all datasets shown in Fig. 3 depicted in the Supplemental online material. Akin series of JMAK kinetic peaks (with *m* ranging between 1 – 4.5) were simulated for different combinations of E and A values listed in Table 2, always at 1 and 10 °C·min^{-1}; the series are denoted with letters A – I. In addition to the non-isothermal heating temperature programs, simulations of the isothermal JMAK kinetics were also performed, with the example being shown in Fig. 3B. Since the rate constant K(T) is factually constant during the isothermal crystallization, its impact on the shape of the isothermal JMAK peaks is largely irrelevant, influencing only the dα·dt^{-1}-t scaling (scaling of X and Y axes) but not the actual shape of the JMAK kinetic peaks. Hence, only two batches of isothermal JMAK peaks (at 100 and 500 °C, with $K = 8{\cdot}10^{-4}$ s^{-1} and $K = 4{\cdot}10^{-4}$ s^{-1}, respectively) were simulated to confirm the results for the non-isothermal heating data.

To further extend the matrix of tested conditions, the JMAK kinetic data based on the non-Arrhenian K(T) dependences were simulated. In Fig. 3C, the Hoffman-Lauritzen nucleation-growth concept paired with the JMAK model (combination of Eqs. 4 and 5) was used to simulate the crystallization during cooling for the following parameters: cooling rate $q^- = 1$ °C·min^{-1}, $\ln(A/s^{-1}) = 10$, $K_g = 100{,}000$ K^2, $T_g = 30$ °C, $T_m = 200$ °C (data typical for fast-crystallizing thermoplastics, such as, e.g., polyamide 12 [68]). Since a low cooling rate was selected for the theoretical modeling, the crystallization proceeds at the onset (high-T)

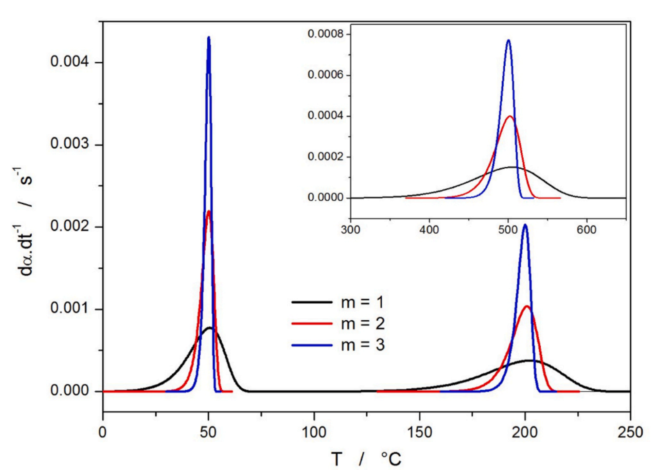

Fig. 2. Three temperature regions for which three batches of JMAK peaks were simulated and consequently fit by the AC kinetic model.

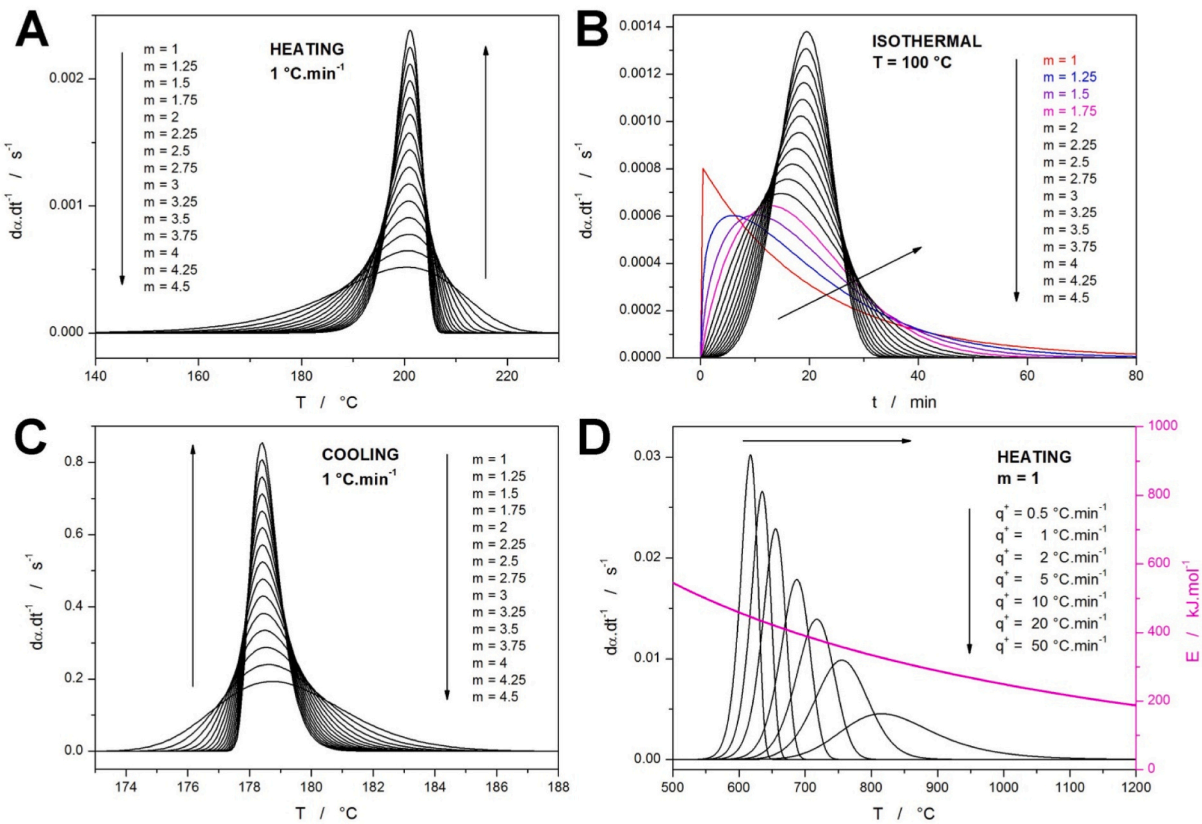

Fig. 3. A) Example set of JMAK kinetic peaks (at 1 °C·min^{-1}) simulated for the heating in the mid-temperature range, Arrhenius K(T), and the full span of the kinetic exponents m. B) Example set of JMAK kinetic peaks simulated for the isotherm at 100 °C and full span of the kinetic exponents m. C) Example set of JMAK kinetic peaks simulated for the cooling in the mid-temperature range, Hoffman-Lauritzen K(T), and the full span of the kinetic exponents m.D) Example set of JMAK kinetic peaks simulated for the heating in the high-temperature range, Arrhenian $K(T)$ with temperature-dependent $E(T)$, $m = 1$, and several q$^+$ (ranging over 2 orders of magnitude). The magenta line and right axis show the E(T) dependence.

Table 2
Non-isothermal JMAK data series simulated with Arrhenius K(T) characterized by the following E and A values; the temperature column indicates a rough temperature region in which the maximum crystallization rate is reached.

series	E / kJ·mol^{-1}	ln(A/s^{-1})	T / °C
A	50	11.7	50
B	50	5.05	200
C	50	−0.8	500
D	150	50	50
E	150	31.5	200
F	150	15.75	500
G	300	106.5	50
H	300	70.4	200
I	300	39.8	500

edge of the K(T)-T dependence – see the Supplemental online material. Hence, the data exhibit the typical JMAK asymmetry – reversed in temperature but still with fingerprint asymmetry with respect to the skewing to higher α values – as can be seen in comparison with the data depicted in Fig. 3A.

The ultimate test for the validity of the JMAK-AC correlation was prepared by simulating the JMAK kinetic data based on another non-monotonous K(T) dependence, but this time with the non-isothermal (heating) crystallization proceeding at higher q$^+$ near/above the inflection point of the onset (low-T) edge of the K(T)-T dependence – see the Supplemental online material. Such mutual position of the crystallization and K(T) peaks leads to a deformation of the shape of the JMAK kinetic peak – as can be seen in Fig. 3D [36]. Note that the non-monotonous K(T) dependence was simulated using Eq. 3 with the following temperature dependences of E and A:

$$E(T) = 1000 \bullet R \bullet \left(-70 \bullet \frac{1}{T} + 25 \right) \tag{8}$$

$$\log A(T) = 45.673 \bullet \frac{1}{T} + 28.628 \tag{9}$$

with T being introduced in Kelvin. Note that the values are loosely based on the data for the crystallization of the CeO_2-doped Bioglass 45S5 [41] (more details in [41]).

As premised in the opening paragraph of this section, the different series of JMAK kinetic peaks (introduced in Fig. 3) were simulated to be consequently fit by the AC kinetic model. To preserve a maximum correspondence between the two sets of equations (Eq. 1+5 vs. Eq. 1+6), the identical temperature dependence of the rate constants, as were input into the initial JMAK simulations, were used during the curve-fitting by the AC model. The only quantities variable during the curve-fitting were the kinetic exponents M and N and the pre-exponential factor A. In Fig. 4A, the values of M, N, and $\Delta\ln(A/s^{-1})$ defined as "$\Delta\ln A = \ln A_{AC} - \ln A_{JMAK}$" determined from the curve-fitting are displayed in dependence on the JMAK kinetic exponent m; the correlation coefficients of the fits were always $r > 0.999999$. For the non-isothermal data, the variations in the individual quantities are displayed – the error bars (much smaller than the magnitude of the points) correspond to the standard deviations calculated from the results obtained for the 9 data series listed in Table 2. To increase the density of the mapped JMAK-AC correlations respective to the value of the kinetic exponent m, the isothermal series of the JMAK peaks were simulated with the step of $\Delta m = 0.2$, starting at the m = 1 value and finishing with m = 4.4. Identical

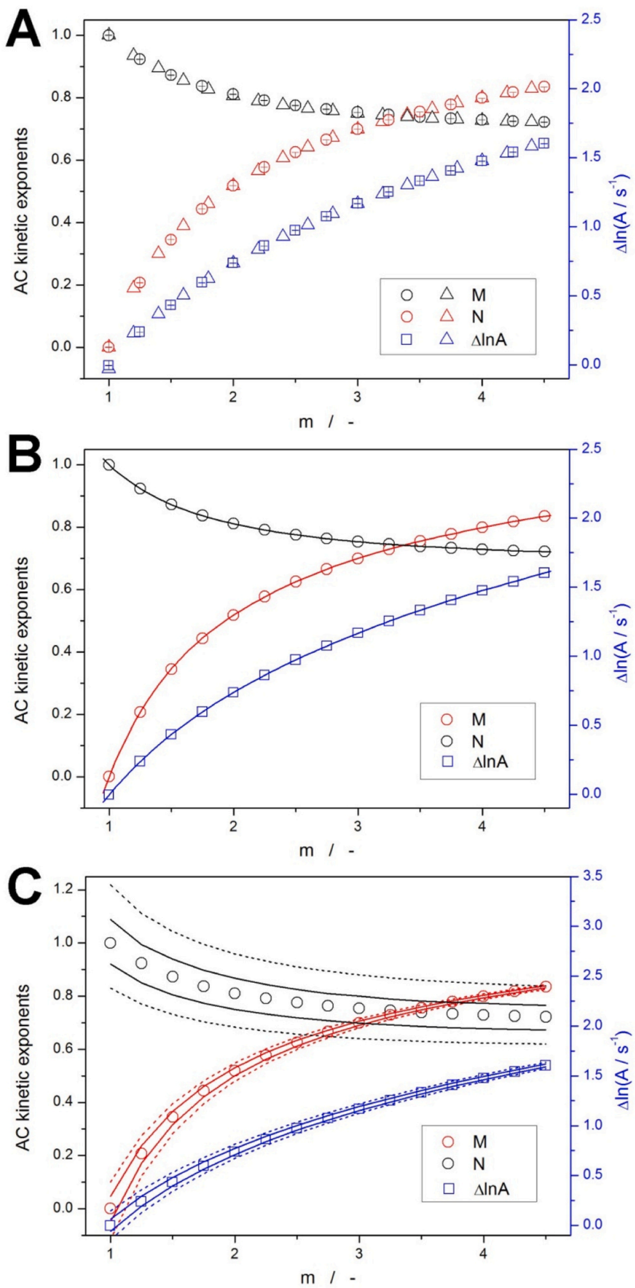

Fig. 4. A) AC kinetic exponents (left axis) and ΔlnA (right axis) obtained by fitting the JMAK kinetic peaks simulated for different m by the AC model (with identical K(T)). Circles and squares indicate the mean values obtained from the non-isothermal simulations; triangles correspond to the results from the isothermal simulations.B) The non-isothermal data from graph A fit by 6th-order polynomial functions.C) The confidence intervals (solid lines for r = 0.999 and dotted lines for r = 0.995) for the fits shown in graph B.

JMAK-AC correlations (expressed through the M, N, and lnA quantities obtained from the curve-fitting) were also obtained for the JMAK model paired with the non-Arrhenian K(T) dependences, i.e., for the data shown in Figs. 3C and 3D.

The three dependencies depicted in Fig. 4A were fit by the sixth-order polynomial functions with r > 0.99999; see Fig. 4B. The parameters/coefficients of these fits are listed in Table 3, denoted in accordance with the following equation:

$$Q_{AC} = P_0 + P_1 \bullet m + P_2 \bullet m^2 + P_3 \bullet m^3 + P_4 \bullet m^4 + P_5 \bullet m^5 + P_6 \bullet m^6 \tag{10}$$

Table 3
Coefficients of Eq. 10 fitting the data in Fig. 4B.

P_x / Q_{AC}	M	N	$\Delta\ln(A/s^{-1})$
P_0	−2.483928	1.905756	−1.722789
P_1	4.871674	−1.762338	2.618086
P_2	−3.561976	1.271748	−1.186480
P_3	1.487950	−0.525210	0.333611
P_4	−0.356124	0.125048	−0.049652
P_5	0.045459	−0.015937	0.002997
P_6	−0.002398	0.000841	0

where Q_{AC} is the respective quantity relevant for the AC model (M, N, or ΔlnA), and P_n are the constants of the polynomial function. These are the master curves, expressing the universal correlation between the JMAK and AC model (Eqs. 5 and 6), the validity of which has been confirmed for all circumstances. It is worth noting that the increase of ΔlnA with increasing *m* is a consequence of the specific autocatalytic function of the kinetic exponent M, which (as was shown in Fig. 1C) leads to the shift of the kinetic peaks to higher T (or generally, to higher crystallization times). This effect then needs to be compensated by the increase of the pre-exponential factor A_{AC} so that the AC peak is identical to the JMAK peak.

In addition to the master curves, the confidential intervals were also calculated, indicating the r = 0.999 and r = 0.995 correlations of the AC model with the JMAK kinetics – see Fig. 4C. The intervals were always determined for the particular AC quantity (M, N, ΔlnA) being changed, and the other two AC quantities being fixed at the values exactly corresponding to the r = 1 correlation with the JMAK data (that are characterized by m and $\ln A_{JMAK}$). As such, the confidential intervals for the ΔlnA quantity can be ignored since they can be (and would be) compensated during the real-life non-linear optimization by the $\ln A_{AC}$ quantity, which does not have its interpretation set with regard to the absolute value of *A*. As is apparent, the AC-JMAK correlation tolerance is almost an order of magnitude higher for the kinetic exponent N compared to the value of M. In addition, the absolute range of the N values is significantly more limited in comparison with that of the M values (0.62 – 1.22 for N vs. 0 – 0.84 for M). This implies that for the interpretation of the potential correspondence or deviations of the real-life data from the JMAK behavior, the value of N should be used to set/estimate the base degree of correspondence with the JMAK kinetics, and the value of M should be used to eventually approximate the value of the JMAK exponent m (if N falls within the 0.62 – 1.22 range). For N values outside of the r ≥ 0.995 correlation, the value of M is largely irrelevant, and the interpretation should be primarily based on the explanation of the change of asymmetry dictated by the value of N. Several examples of such interpretations will be given in Section 3.2.

3. Discussion

The discussion of the obtained results will be split into two sub-sections. In the first sub-section, the relevance and accuracy of the kinetic predictions made on the basis of the AC model being correlated with the JMAK kinetics will be demonstrated. The second sub-section will introduce concrete examples of the interpretation of the AC kinetics in terms of the nucleation-growth concept; archetypal cases based on different combinations of the M and N kinetic exponents will be focused.

3.1. Kinetic predictions of the AC description imitating JMAK kinetics

The ability to extrapolate the kinetic information obtained from the experimentally measured data to situations/conditions for which the experiment is impossible or at least difficult (e.g., exceedingly long) to perform is extremely valuable and makes the kinetic predictions the main reason for the ongoing development of the methodologies of kinetic analysis (KA). In the present case, the kinetic predictions can be used to verify the practical implementability of the correlations depicted in Fig. 4C because the extrapolation beyond the measured data range is not only the primary goal of KA, but it can also amplify certain types of data errors/inaccuracies. As such, the comparison of the kinetic predictions obtained for different datasets represents an ultimately beneficial way of considering their equality.

For the present data, the kinetic predictions for the isothermal annealing (crystallization) at 500 °C were calculated to approximate the temperature range of interest for classical glass science. Note, however, that identical relations and conclusions would also be obtained for any other temperature range. Thus, the kinetic predictions were based on the last entry from Table 2, i.e., $E = 300$ kJ·mol^{-1} and $\ln(A/s^{-1}) = 39.8$. The usability of the AC correlations depicted in Fig. 4C was considered for the two most important cases of the nucleation-growth behavior when $m_{JMAK} = 1$ and $m_{JMAK} = 3$. The predictions of the kinetic (isothermal) behavior was calculated for the true JMAK model and for the individual boundaries determined for M, N, and ΔlnA, representing $r = 0.995$ (dotted lines in Fig. 4C).

The prediction of the crystallization extent during annealing at 500 °C is for the JMAK kinetics with $m = 1$ and for the AC kinetics approximating the true JMAK model with the 0.995 correlation shown in Fig. 5A. Note that the simulations of the AC kinetics were always performed for a single change of one parameter (M, N or ΔlnA), while the other ones were preserved at values defined by the exact correspondence with the JMAK kinetics. For example, the curve marked as "M –" corresponds to the simulation with $M = -0.117$, $N = 1$, $\Delta\ln A = 0$, where the M value represents the low $r = 0.995$ boundary for this parameter at $m = 1$ from Fig. 4C and the other two parameters are at values represented in Fig. 4C by the points at $m = 1$. Interestingly, for low α (beginning of the crystallization process), the largest errors are caused by the changes of parameter M, whereas at high α values, the largest difference from the JMAK kinetics is caused by the deviation of ΔlnA and (even more so) N. These findings can give guidelines for maximizing the accuracy of the practical kinetic predictions utilizing the JMAK-AC correlation. Nevertheless, overall, even for the largest deviations, the "error" in the α determination is ∼ 5 %, which is perfectly acceptable for any practical application of glass-ceramics development. The analogous kinetic predictions calculated for the JMAK exponent $m = 3$ are shown in Fig. 5B. Here, owing to the generally narrower kinetic peaks manifesting at higher *m* values, the α "errors" associated with the $r = 0.995$ deviations of the AC kinetics are less than 5 %. Otherwise, similar conclusions are valid regarding the dominant influences of the M and N exponents over the particular parts of the crystallization process. In conclusion, the $r = 0.995$ correspondence between the AC and JMAK kinetics depicted in Fig. 4C can be indeed utilized as universal practical limits for the description as well as the implementation of the AC imitation of the nucleation-growth theory.

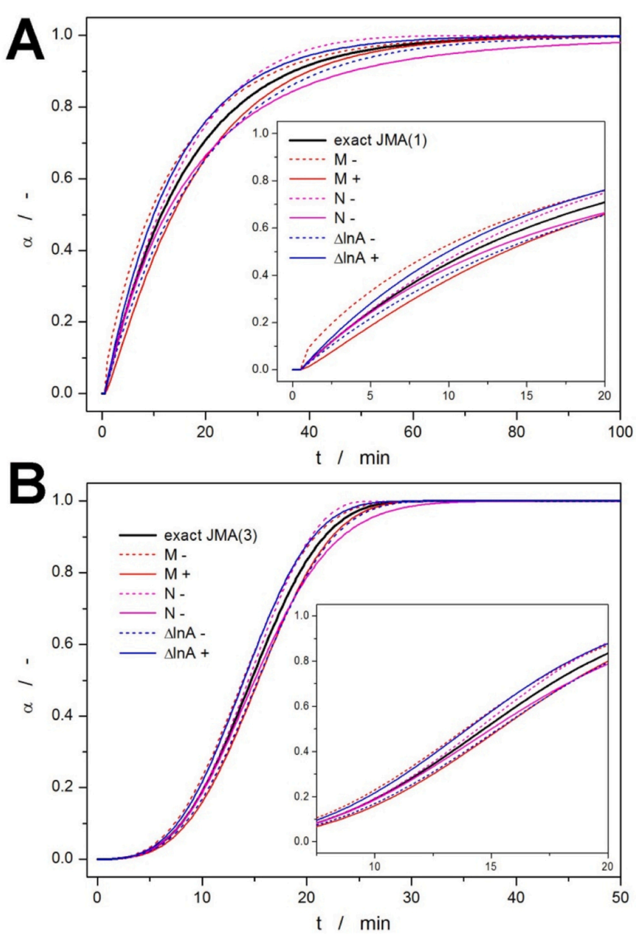

Fig. 5. The predictions of the kinetic behavior during annealing at 500 °C. The predictions are simulated for $E = 300$ kJ·mol^{-1} and $\ln(A/s^{-1}) = 39.8$. The parameters M, N, and ΔlnA were taken as maximum boundaries for the AC-JMAK correlation with $r = 0.995$ (dotted lines in Fig. 4C). Always only one deviated parameter was used (either its lower or upper value, denoted "-" and "+", respectively), the other two parameters were input into the simulation as values exactly corresponding to the JMAK kinetics. The exploration of the impact of the deviated M, N, and ΔlnA parameters was done for the base kinetics corresponding to $m = 1$ (graph A) and $m = 3$ (graph B).

3.2. Interpretation of the AC kinetics in terms of the nucleation-growth concept

In this section, the archetypal deviations of the AC kinetics from the JMAK model will be shown, and their interpretation in terms of the nucleation-growth concept will be introduced. Note that the degree of the deviations of the M and N exponents from the values corresponding to the JMAK kinetics can be then used to identify the main physicochemical origin of the deviations (with the main advantage of the AC model being the split of the individual kinetic effects among the two kinetic exponents – as will be described below). First, the deviations of only one of the AC parameters will be interpreted – the archetypal variants coming into question are shown in Fig. 6; the alterations originate from two initial states corresponding to the JMAK kinetics with $m = 1$ (identical to $M = 0$ & $N = 1$; Figs. 6A and 6C) and $m = 3$ (identical to $M = 0.6995$ & $N = 0.7534$; Figs. 6B and 6D). The increase of the exponent M is primarily associated with delayed crystal growth (mathematically compensated via the increased value of the pre-exponential factor A), where the fingerprint of JMAK's asymmetry only very slightly increases (becomes less negative) with increasing M. The delayed onset of the crystallization process is then partially balanced by the faster (autocatalyzed) course of the process (once it is initiated) so that the peak width (temperature range in which the process occurs) is largely reduced. All these effects increase in magnitude/rapidity with increasing JMAK exponent *m*, as evidenced by the comparison of Figs. 6A and 6B. With regard to the physicochemical interpretation of this effect, the crystallization onset (either earlier or delayed) can be influenced, e.g., by the presence of pre-existing nuclei or crystalline phase in general, by the presence of impurities, nanoparticles, or phase-separated domains, by the presence of moisture with plasticizing effect, by the products of oxidative or decomposing reactions, by internal or external mechanical stress, by exposure to radiation, electrical or magnetic fields, or by effects affecting the material's surface or volume mobility (self-diffusion) – including the structural relaxation [69–75].

Contrary to the AC exponent M, the value of the exponent N does not influence the onset edge of the kinetic peak at all – see Figs. 6C and 6D. Instead, the increasing N leads to a prolonged endset tail of the kinetic peak, largely changing the peak asymmetry toward higher values – from negative to positive (causing larger skewing to low T and α values). The

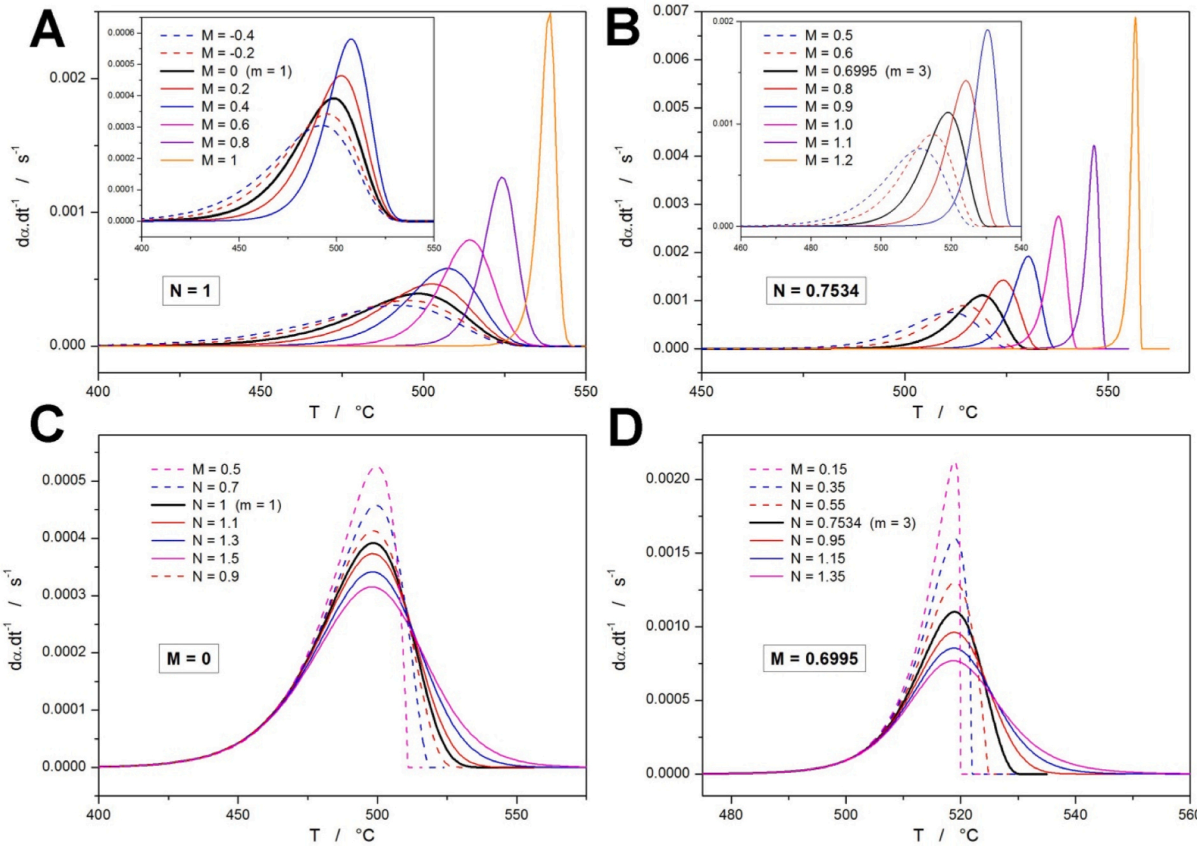

Fig. 6. Series of AC kinetic peaks simulated for Arrhenius K(T) with E = 300 kJ·mol^{-1} and ln(A/s^{-1}) = 39.8. Graphs monitor changes in the peak shape and position with the change of one of the AC kinetic exponents; the base behavior was simulated either for $m = 1$ (graphs A and C) or $m = 3$ (graphs B and D). The insets show the data range zoomed-in on the peaks simulated with low M values.

low N values (< 0.65) can essentially represent the transition from the nucleation-growth crystallization mechanism to the zero-order crystallization kinetics (akin to, e.g., the contracting-sphere kinetics), typical for the surface crystallization from a large number of nuclei, where a continuous layer of crystalline material forms on the surface, and the growth front propagates at a constant rate inwards. The values of N > 1.2 then result in a larger and larger portion of the crystallization proceeding during the final stages of the overall transformation process (which manifests as a prolonged endset peak tail). The physicochemical reason for such behavior may be associated with, e.g., atomic/molecular mobility being constrained due to the spatial restrictions, density changes, or internal stresses arising from the preceding formation of the crystalline phase, which can slow down the consequent crystal growth occurring in the late process stages within the isolated amorphous regions. Another cause for the prolonged crystallization can be a broad quality distribution of nucleation sites (differing in energy barrier, i.e., activation energy, and being represented by phasic, compositional, or polymorphic inhomogeneities), which can result in staggered crystal growth. One of the very important mechanisms leading to a prolonged crystallization tail is also the secondary crystallization being initiated from newly formed active sites at an already existing crystalline surface [69–75]. Similar to the impact of exponent M, the influence of the N also increases with the increasing corresponding JMAK exponent m (based on which the AC description was derived in the first place).

Whereas in Fig. 6, the sole impact of each AC kinetic exponent was tested separately, joint influences of both exponents are shown in Fig. 7 – Figs. 7A and 7C are based on the correspondence between the JMAK kinetics with m = 1 (identical to M = 0 & N = 1), and Figs. 7B and 7D were derived based on the correspondence with JMAK kinetics with m = 3 (identical to M = 0.6995 & N = 0.7534). Starting with similar trends for the M and N exponents (M increasing while N increasing, and vice-versa), as depicted in Figs. 7A and 7B for $m \approx 1$ and $m \approx 3$, respectively, the changes in the shape of the crystallization peak associated with each of the AC exponents partially compensate for each other. This is particularly apparent in the case of the crystallization peak being narrowed by increasing M and, at the same time, being widened by increasing N. On the contrary, both AC exponents impact (with regard to their trends) the asymmetry of the crystallization peak in a similar fashion, i.e., the individual increases of M as well as of N lead to the shift of the peak asymmetry to more positive values. Note, however, that the exponent N has a much stronger impact on the peak asymmetry than the exponent M. In conclusion, the similarly deviating M&N (compared to their fingerprint values indicating equality with the JMAK kinetics) result in more-or-less similarly wide and high crystallization peaks with markedly shifted asymmetry.

The opposite case, i.e., the trends in M and N being opposite to each other (M increasing while N decreasing and vice-versa), is depicted in Figs. 7C and 7D. Here, the M&N trends both similarly boost the changes in width and height of the crystallization peak, resulting in markedly changed rapidity of the crystal growth. On the other hand, the changes in asymmetry (dominantly dictated by N) are slightly compensated by the mathematical impact of the autocatalytic exponent M. Note that in either case, the impacts of M&N on the actual temperature position of the crystallization peak do not have to be considered for the interpretation of the real-life data, because its potential compensation by the pre-exponential factor A can be expected (but cannot be determined).

4. Conclusions

Theoretical simulations were used to assess the correlation between

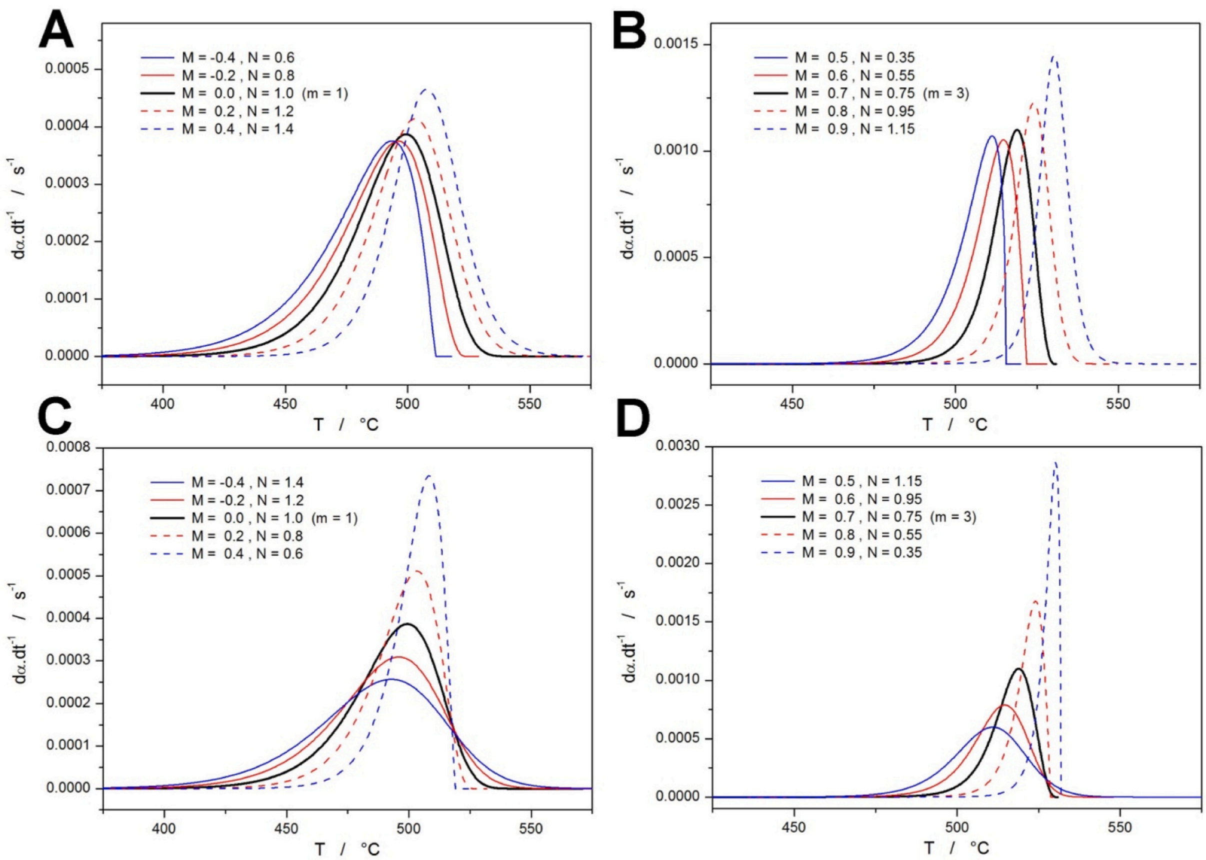

Fig. 7. Series of AC kinetic peaks simulated for Arrhenius K(T) with E = 300 kJ·mol^{-1} and ln(A/s^{-1}) = 39.8. Graphs monitor changes in the peak shape and position with the simultaneous changes of both AC kinetic exponents; the base behavior was simulated either for $m = 1$ (graphs A and C) or $m = 3$ (graphs B and D).

the JMAK and AC kinetics. It was found that, indeed, unequivocal reciprocity exists between the two crystallization models, where a unique combination of the M+N + ΔlnA AC parameters corresponds to each value of the JMAK kinetic exponent m. The functions expressing these AC parameters (as the dependent variables) in dependence on m can be modeled by the nth-order polynomial functions (listed in Table 3), where for n = 5 – 6, the correlation coefficients r > 0.99999 were achieved. This correlation holds under any circumstances, assuming the validity of the base solid-state kinetic equation: dα/dt = K(T)·f(α), as was verified for isothermal conditions as well as non-isothermal conditions, with the rate constant K(T) expressed via Arrhenius function, Hoffman-Lauritzen function, and also an arbitrary Arrhenius function with temperature-dependent E(T) and A(T). The confidence intervals for the JMAK-AC correlation show that the N kinetic exponent can deviate by ± 0.1 – 0.2 (depending on the corresponding m) to still fall within the r > 0.995 correlation with the JMAK kinetics. For the kinetic exponent M, the allowed deviations are much smaller: between 0.02 and 0.1. It was further shown that the kinetic predictions calculated with the M, N, and ΔlnA parameters deviating within these confidence intervals are still very accurate, sufficiently imitating the true JMAK kinetics with any practical purpose in mind, e.g., for adjusting temperature programs intended for the preparation of the glass-ceramics.

Most importantly, for the larger M and/or N deviations from the values corresponding to the JMAK kinetics, the individual impact of each exponent was identified, and the physico-chemical interpretations of the associated distortions of the crystallization peak were suggested. These lists attribute the first physically meaningful interpretation to the autocatalytic Šesták-Berggren model (otherwise an empirical, purely mathematic function) when used for the description of the crystallization from the glassy matrix. Apart from the confirmation of the correspondence between experimental data and the formal nucleation-growth kinetics (the actual confirmation should still be based on the complementary microscopic observation), the present findings can be further utilized as a basis for attribution of certain physicochemically modeled types of deviations from the JMAK behavior to the characteristic sets of the AC parameters. Ultimately, with the extended theoretical basis, the AC model could potentially represent a single universal and physically meaningful kinetic model for all types of crystal growth behavior.

CRediT authorship contribution statement

Roman Svoboda: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jeurceramsoc.2025.117716.

References

[1] Y. Okutan, B. Kandemir, M. Dönmez, M. Yücel, Effect of the thickness of cad-cam materials on the shear bond strength of light-polymerized resin cement, Eur. J. Oral. Sci. 130 (2022) e12892.
[2] H. Wu, W. Teng, X. Feng, J. Li, Preparation and properties of the glass-ceramics from low ti-bearing blast furnace slag and waste glass, Int. J. Appl. Ceram. Technol. 21 (2023) 844–854.
[3] Y. Jiang, C. Zhang, J. Xu, J. Xu, X. Zhan, N. Ma, F.-C. Tsai, An overview of dental glass–ceramics: from material design to the manufacturing process, Int. J. Cer. Eng. Sci. 6 (2024) e10224.
[4] L. Chen, T. Sun, Z. Hao, J. Xie, J. Zhang, J. Han, M. Wang, X. Cui, X. Liang, Ultrahigh hardness Li_2O–MgO–Al_2O_3–SiO_2 glass–ceramics containing multiphase nanocrystals, J. Am. Ceram. Soc. 105 (2022) 7614–7624.
[5] P. Li, Y. Zhi, L. Wang, Y. Xu, P. Lu, J. Zhang, Advanced transparent glass-ceramics via laser anisotropic nanocrystallization, Laser Photonics Rev. 18 (2024) 2301403.
[6] A. Naumov, G. Shakhgil'dyan, N. Golubev, A. Lipatiev, C Fedotov, R. Alekseevet, E. S. Ingateva, V.I. Savinkov, V.N. Sigaev, Tuning the coefficient of thermal expansion of transparent lithium aluminosilicate glass-ceramics by a two-stage heat treatment, Ceramics 7 (2023) 1–14.
[7] M. Li, C. Dong, Y. Ma, H. Jiang, Light-transmitting lithium aluminosilicate glass-ceramics with excellent mechanical properties based on cluster model design, Nanomaterials 13 (2023) 530.
[8] Y. Sun, Z. Yang, D. Cai, Z. Zhang, Q. Liu, S. Fang, L. Feng, L. Shi, Y. Wang, D. Jia, Crystallization kinetics, properties of α-cordierite based glass-ceramics prepared by glass powder sintering, J. Inorg. Mat. 37 (2022) 1351.
[9] A. Shearer, M. Montazerian, B. Deng, J. Sly, J. Mauro, Zirconia-containing glass-ceramics: from nucleating agent to primary crystalline phase, Int. J. Ceram. Eng. Sci. 6 (2024) e10200.
[10] B. Kettlewell, D. Boyd, State of the art review for titanium fluorine glasses and glass ceramics, Materials 17 (2024) 1403.
[11] M. Backhaus-Ricoult, M. Comte, E. François, A. Rezikyan, B. Wheaton, Evolution of electrical, structural, and chemical properties of li-aluminosilicate glass during crystallization, J. Am. Ceram. Soc. 107 (2023) 897–908.
[12] Y. Yang, X. Feng, Y. Huang, J. Wang, L. Yuan, Q. Zhang, Q. Fu, S. Zhou, Effect of alkaline earth metal ions R^{2+} (R = Mg, Ca, Sr, and Ba) on the properties of $LiNbO_3$ glass ceramics, J. Am. Ceram. Soc. 108 (2025) e20397.
[13] P. Rao, R. Singh, Sintering and thermal expansion behaviors of glass and glass–YSZ composites as self-repairable seals for sofc, J. Am. Ceram. Soc. 106 (2022) 157–165.
[14] Y. Guo, Y. Du, G. Wang, Y. Xu, H. Zhang, L. Deng, H. Chen, M. Zhao, The direct experimental observation and solidification mechanism of heavy metals in blast furnace slag glass-ceramics, Int. J. Appl. Ceram. Technol. 21 (2023) 796–805.
[15] L. Zeng, H. Sun, T. Peng, Effect of borax on sintering kinetics, microstructure and mechanical properties of porous glass-ceramics from coal fly ash by direct overfiring, Front. Chem. 10 (2022) 839680.
[16] Z. Pei, H. Huang, X. Guo, J. Zhang, M. Chen, Preparation and properties of MgO-Al_2O_3-SiO_2 glass–ceramics with controllable crystalline phases, Crystals 13 (2023) 1261.
[17] N. Possolli, H. Modolon, H. Souza, C. Pich, O. Montedo, E. Angioletto, S. Arcaro, Antimicrobial characteristics of the Li_2O–ZrO_2–SiO_2 bioactive glass and glass-ceramic for bone regeneration and wound healing, Int. J. Appl. Ceram. Technol. 21 (2024) 2567–2577.
[18] D. Arıbuğa, U. Akkaşoğlu, B. Çiçek, Ö. Balcı, Enhanced sinterability, thermal conductivity and dielectric constant of glass-ceramics with PVA and BN additions, Materials 15 (2022) 1685.
[19] P. Odewole, Properties of glass-ceramics foam based on granite dust-clay-maize cob composite as a sustainable building materiál, J. Sustain. Constr. Mater. Technol. 7 (2022) 1–8.
[20] N. Drancourt, E. Nicolas, J. Veyrune, M. Bessadet, Comprehensive cad/cam prosthetic rehabilitation management in a young patient with agenesis: a case report, Healthcare 10 (2022) 382.
[21] G. Guzman-Perez, C. Jurado, S. Alresayes, F. Floriani, S. Rojas-Rueda, A. Tsujimoto, Navigating esthetic challenges: immediate implant placement and comprehensive restorative solutions—a clinical case with 2-year follow-up study, Case Rep. Dent. 1 (2024) 1186299.
[22] Y. Zhang, S. Vardhaman, C. Rodrigues, B. Lawn, A critical review of dental lithia-based glass–ceramics, J. Dent. Res. 102 (2023) 245–253.
[23] M. Kolarova, Mechanical characterisation of cad/cam dental material and natural teeth, Ceram. Silik. 68 (2024) 516–526.
[24] E. Elsharkawy, A. El-Kouedi, T. Shokry, Three-dimensional assessment of internal adaptation measurement of three cad/cam ceramic systems, Braz. Dent. J. 33 (2022) 121–128.
[25] Z. Tong, J. Sun, S. Liu, W. Zhang, M. Kuang, Study on a new process and its kinetics of iron recovery and glass-ceramics preparation from desulfurization slag, J. Min. Metall. B 58 (2022) 169–178.
[26] N. Kulrat, C. Busabok, S. Intarasiri, S. Dangtip, W. Khongwong, Fabrication of glass-ceramics composite by infiltration of lithium tetraborate glass into porous magnesium aluminate spinel ceramic, J. Met. Mater. Miner. 33 (2023) 89–94.
[27] H. Li, H. Zou, Z. Sun, Y. Xu, C. Wang, X. Xie, J. Yi, F. Zhao, Spectroscopic and surface crystallization characterizations of yttrium-doped phosphate glasses, Crystals 12 (2022) 109.
[28] K. Shvydyuk, J. Nunes-Pereira, F. Rodrigues, A. Silva, Review of ceramic composites in aeronautics and aerospace: a multifunctional approach for tps, tbc and dbd applications, Ceramics 6 (2023) 195–230.
[29] L. Yang, D. Wang, Z. Guo-xiang, Z. Lan, Z. Yang, Glass-ceramic coating on silver electrode surface via 3d printing, Materials 16 (2023) 3276.
[30] E. Schwarzer-Fischer, J. Abel, J. Sieder-Katzmann, M. Propst, C. Bach, U. Scheithauer, A. Michaelis, Study on ceramfacturing of novel alumina aerospike nozzles by lithography-based ceramic vat photopolymerization (ceram vpp), Materials 15 (2022) 3279.
[31] T. Honma, K. Maeda, S. Nakane, K. Shinozaki, Unique properties and potential of glass-ceramics, J. Ceram. Soc. Jpn. 130 (2022) 545–551.
[32] M. Li, C. Dong, Y. Ma, H. Jiang, Light-transmitting lithium aluminosilicate glass-ceramics with excellent mechanical properties based on cluster model design, Nanomaterials 13 (2023) 530.
[33] A. DeCeanne, C. Wilkinson, M. Dittmer, C. Ritzberger, M. Rampf, J. Mauro, Experimental analysis and machine learning modeling of optical properties of lithium disilicate glass–ceramics comprising lithium tantalate as a secondary phase, Int. J. Appl. Glass Sci. 13 (2022) 720–728.
[34] J. Šesták, Science of Heat and Thermophysical Studies: a Generalized Approach to Thermal Analysis, Elsevier, Amsterdam, 2005.
[35] P. Holba, J. Šesták, Imperfections of Kissinger evaluation method and crystallization kinetics, Glass Phys. Chem. 40 (2014) 486–495.
[36] R. Svoboda, Johnson-Mehl-Avrami kinetics as a universal description of crystallization in glasses? J. Eur. Ceram. Soc. 44 (2024) 4064–4082.
[37] R. Svoboda, J. Chovanec, S. Slang, L. Beneš, P. Konrád, Single-curve multivariate kinetic analysis: application to the crystallization of commercial Fe-Si-Cr-B amorphous alloys, J. Alloy. Compd. 889 (2022) 161672.
[38] D. Brandová, R. Svoboda, Z.Olmrová Zmrhalová, J. Chovanec, R. Bulánek, Crystallization kinetics of glassy materials: the ultimate complexity? J. Therm. Anal. Calor. 134 (2018) 825–834.
[39] R. Svoboda, J. Prikryl, A.V. Kolobov, M. Krbal, Preparation of Sb_2Se_3-based ceramics and glass-ceramics from native thin films deposited on Kapton foil, Ceram. Int. 48 (2022) 17065–17075.
[40] R. Svoboda, J. Přikryl, V. Čičmancová, Vít Prokop, A.V. Kolobov, M. Krbal, Crystal growth in amorphous selenium thin films – reviewed and revisited: direct comparison of microscopic and calorimetric measurements, Cryst. Growth Des. 21 (2021) 7087–7097.
[41] M. Chromčíková, R. Svoboda, B. Hruška, B. Pecušová, A. Nowicka, Thermo-kinetic and structural characterization of Ce-doped glasses based on Bioglass 45S5, Mater. Chem. Phys. 304 (2023) 127833.
[42] J.D. Hoffman, G.T. Davis, J.I. Lauritzen Jr., The rate of crystallization of linear polymers with chain folding, in: in: Treatise on Solid State Chemistry, 3, Springer, Boston, MA, 1976, pp. 497–614.
[43] S. Cheng, B. Lotz, Enthalpic and entropic origins of nucleation barriers during polymer crystallization: the Hoffman–Lauritzen theory and beyond, Polymer 46 (2005) 8662–8681.
[44] W.A. Johnson, R.F. Mehl, Reaction kinetics in process of nucleation and growth, Trans. Am. Inst. Min. Metall. Eng. 135 (1939) 416–442.
[45] M. Avrami, Kinetics of phase change. I General theory, J. Chem. Phys. 7 (1939) 1103–1112.
[46] M. Avrami, Kinetics of phase change. II transformation-time relations for random distribution of nuclei, J. Chem. Phys. 8 (1940) 212–224.
[47] M. Avrami, Granulation, phase change, and microstructure kinetics of phase change. III, J. Chem. Phys. 9 (1931) 177–184.
[48] A.N. Kolmogorov, On the statistical theory of the crystallization of metals, Bull. Acad. Sci. URSS (Cl. Sci. Math. Nat.) 1 (1937) 355–359.
[49] J. Šesták, Thermophysical Properties of Solids, Their Measurements and Theoretical Analysis, Elsevier, Amsterdam, 1984.
[50] T. Huang, C. Zhou, Surface diffusion-controlled jonhson–mehl–avrami–kolmogorov model for hydrogenation of Mg-based alloys, J. Phys. Chem. C 127 (2023) 13900–13910.
[51] R. Svoboda, Crystallization of glasses – when to use the Johnson-Mehl-Avrami kinetics? J. Eur. Ceram. Soc. 41 (2021) 7862–7867.
[52] P.W. Atkins, Atkins' Physical Chemistry, W.H. Freeman, New York, 2006.
[53] G. Luciano, R. Svoboda, Simulation and non-linear optimization of kinetic models for solid-state processes, Model. Simul. Mater. Sci. Eng. 32 (2024) 035014.
[54] X. Han, K. Dai, K. Kawakami, Influence of nucleation on relaxation, molecular cooperativity, and physical stability of celecoxib glass, Mol. Pharm. 21 (2024) 1794–1803.
[55] T. Konishi, K. Taguchi, K. Fukao, N. Takagi, Y. Miyamoto, Crystallization with nodular aggregation near the glass transition temperature for syndiotactic polypropylene, ACS Macro Lett. 12 (2023) 208–214.
[56] D. Nie, X. Yin, Z. Cai, J. Wang, Effect of crystallization on shape memory effect of poly(lactic acid), Polymers 14 (2022) 1569.
[57] L. Zhu, J. Jona, K. Nagapudi, T. Wu, Fast surface crystallization of amorphous griseofulvin below Tg, Pharm. Res. 27 (2010) 1558–1567.
[58] D. Musumeci, M. Hasebe, L. Yu, Crystallization of organic glasses: how does liquid flow damage surface crystal growth? Cryst. Growth Des. 16 (2016) 2931–2936.
[59] K. Pach-Zawada, M. Leśniak, K. Filipecka, E. Golis, M. Sitarz, D. Dorosz, J. Filipecki, Tellurite glasses from the $70TeO_2$-5XO-$10P_2O_5$-10ZnO-$5PbF_2$(x= Pb, Bi, Ti) system doped erbium ions—the influence of erbium on the structure and physical properties, Int. J. Mol. Sci. 24 (2023) 3556.
[60] J. Pries, H. Weber, J. Benke, I. Jacob, S. Kaban, M. Wei, P. Wuttig, Lucas. Fragile-to-strong transition in phase-change material $Ge_3Sb_6Te_5$, Adv. Funct. Mater. 32 (2022) 2202714.
[61] L. Zhu, M. Wang, Y. Xu, X. Zhang, P. Lu, Dual effect of ZrO_2 on phase separation and crystallization in Li_2O–Al_2O_3–SiO_2–P_2O_5 glasses, J. Am. Ceram. Soc. 105 (2022) 5698–5710.

[62] Y. Hordieiev, A. Zaichuk, Exploring the impact of Bi_2O_3 addition on the thermal properties and crystallization behavior of lead borosilicate glasses, J. Ovonic Res. 19 (2023) 471–481.

[63] H. Lee, W. Carty, The glass formation boundary in aluminosilicates, Int. J. Appl. Glass Sci. 13 (2022) 708–719.

[64] B. Demirel, M. Erol, Antibacterial borosilicate glass and glass ceramic materials doped with zno for usage in the pharmaceutical industry, ACS Omega 8 (2023) 18735–18742.

[65] S.V. Pershina, Structural, thermal, and dc conductivity properties of WO_3–P_2O_5 glasses doped with B_2O_3, Int. J. Appl. Glass Sci. 14 (2022) 247–255.

[66] Z. He, Y. Gao, J. Ren, J. Chen, J. Qiu, Effect of concentration of Er^{3+} on up-conversion luminescence in $NaYF_4$ containing oxy-fluoride glass-ceramics, J. Am. Ceram. Soc. 107 (2024) 5482–5489.

[67] R. Silveira, G. Hippler, L. Evaristo, S. Buchner, Crystallization in a soda–lime-silicate glass after high pressure, J. Am. Ceram. Soc. 106 (2023) 3410–3417.

[68] F. Paolucci, D. Baeten, P.C. Roozemond, B. Goderis, G.W.M. Peters, Quantification of isothermal crystallization of polyamide 12: modelling of crystallization kinetics and phase composition, Polymer 155 (2018) 187–198.

[69] F. Liu, G. Yang, Effects of anisotropic growth on the deviations from Johnson–Mehl–Avrami kinetics, Acta Mater. 55 (2007) 1629–1639.

[70] T. Huang, C. Zhou, Surface diffusion-controlled Jonhson–Mehl–Avrami–Kolmogorov model for hydrogenation of Mg-based alloys, J. Phys. Chem. C 127 (2023) 13900–13910.

[71] M. Castro, F. Domínguez-Adame, A. Sánchez, T. Rodríguez, Model for crystallization kinetics: deviations from Kolmogorov–Johnson–Mehl–Avrami kinetics, Appl. Phys. Lett. 11 (1999) 2205–2207.

[72] A. Kolb-Telieps, Tan Shu-Song, Investigations to explain the restricted validity of the Avrami-Johnson-Mehl theory for the crystallization of Zr70Ni30, J. NonCryst. Solids 107 (1988) 122–127.

[73] K. Shirzad, C. Viney, Revisiting time-dependent growth and nucleation rates in the Johnson-Mehl-Avrami-Kolmogorov equation, R. Soc. Open Sci. 12 (2025) 241696.

[74] M. Fanfoni, M. Tomellini, Beyond the Kolmogorov Johnson Mehl Avrami kinetics: inclusion of the spatial correlation, Eur. Phys. J. B 34 (2003) 331–341.

[75] J. Farjas, P. Roura, Modification of the Kolmogorov–Johnson–Mehl–Avrami rate equation for non-isothermal experiments and its analytical solution, Acta Mater. 54 (2006) 5573–5579.